

SQA Higher Computing Science

Course Sections and Topics (2025)

1. Software Design and Development

- Development methodologies S-DM-
- Analysis S-A-
- Design S-D-
- Implementation (data types and structures) S-ID-
- Implementation (computational constructs) S-IC-
- Implementation (algorithm specification) S-IA-
- Testing S-T-
- Evaluation S-E-

2. Computer Systems

- Data representation (binary, floating-point, characters) C-D-
- Computer architecture and the fetch–execute cycle C-C
- Performance factors (processor, memory, buses) C-P
- Environmental impact of intelligent systems C-E-
- Security risks, precautions and legislation C-S-

3. Database Design and Development

- Analysis of database problems (purpose, scope, entities, relationships) D-A-
- Entity-relationship diagrams and data modelling D-E-
- Design of tables, attributes, keys and relationships D-D-
- Implementation using SQL (CREATE, INSERT, UPDATE, DELETE, SELECT with joins and functions) D-I-
- Testing and evaluation of database solutions D-T-

4. Web Design and Development

- Analysis of web-based problems W-A-
- Design using wireframes and layout planning W-D-
- Implementation using HTML, CSS and JavaScript W-I-
- Testing (usability, input validation, compatibility) W-T-
- Evaluation (fitness for purpose, usability) W-E-